

HARDENING LINUX SYSTEMS

PRACTICE 1 : PERFORM LUKS (LINUX UNIFIED KEY SETUP) ENCRYPTION

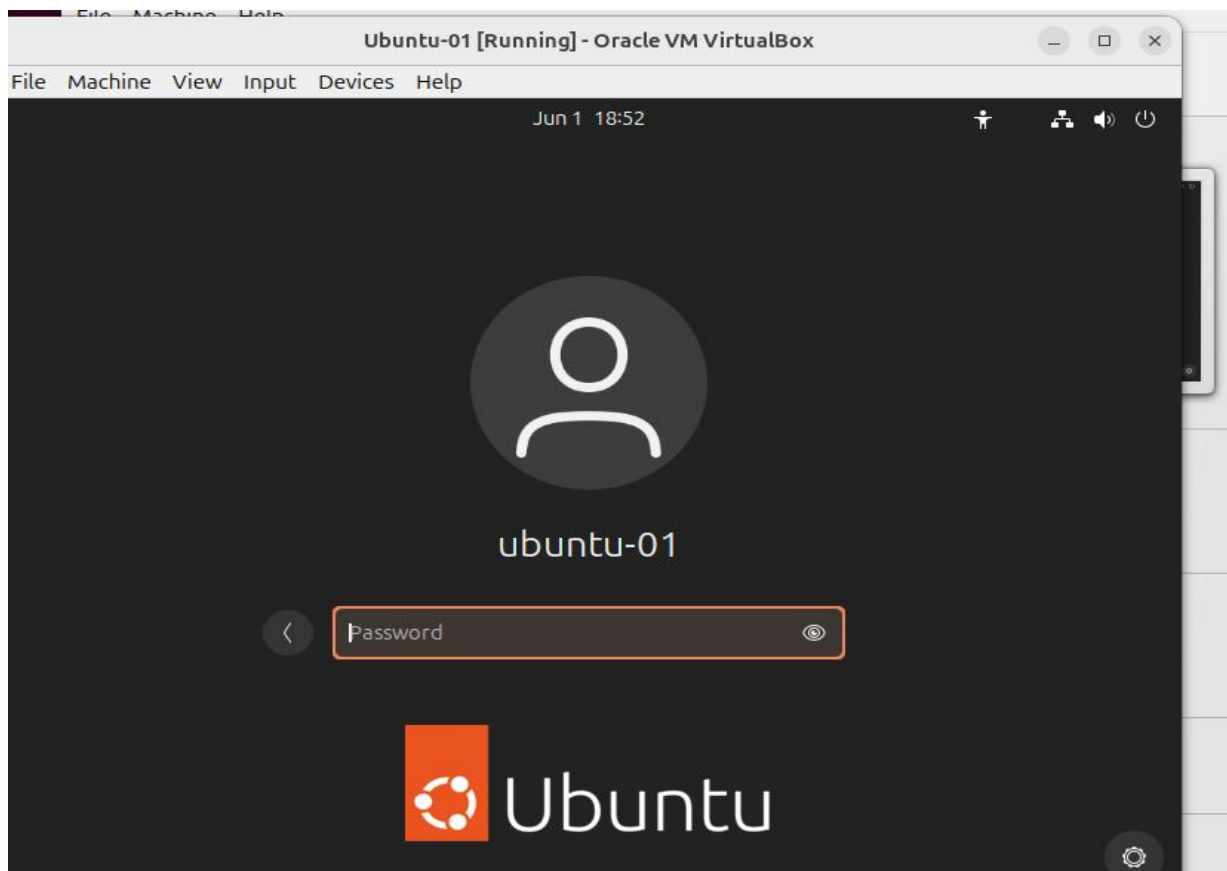
As an administrator, you decide to implement LUKS encryption on the **/dev/sdb** partition of the system to secure the data. For another layer of data security, you decide to use SELinux on the mounted directory where users keep data.

Steps to Perform LUKS Encryption To protect the official data, perform the following steps in the system:

Step 1: Install the cryptsetup tool

- Once ubuntu-01 virtual machines starts and Login with credentials. Install the "cryptsetup" and "parted" packages.

- \$ sudo apt update && sudo apt upgrade -y
- \$ apt-get install -y cryptsetup parted



- \$ sudo parted /dev/sdb -s mkpart primary 0% 100% #OR
- \$ sudo parted /dev/sdb -s mkpart primary 1 -1

```

Error: Expected a partition type.
ubuntu-01@ubuntu-01:~$ sudo parted -s /dev/sdb mkpart primary 0% 100%
ubuntu-01@ubuntu-01:~$

```

Verifications:

parted /dev/sdb-l #List partitions to verify
(Or)

lsblk #Should show /dev/sdb1 as a new partition

```

ubuntu-01@ubuntu-01:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0        7:0      0    4K  1 loop /snap/bare/5
loop1        7:1      0   74M  1 loop /snap/core22/2292
loop2        7:2      0 251.7M  1 loop /snap/firefox/7766
loop3        7:3      0  18.5M  1 loop /snap/firmware-updater/210
loop4        7:4      0 531.4M  1 loop /snap/gnome-42-2204/247
loop5        7:5      0   91.7M  1 loop /snap/gtk-common-themes/1535
loop6        7:6      0   10.8M  1 loop /snap/snap-store/1270
loop7        7:7      0   48.1M  1 loop /snap/snapd/25935
loop8        7:8      0    576K  1 loop /snap/snapd-desktop-integration/343
loop9        7:9      0   49.3M  1 loop /snap/snapd/26865
loop10       7:10     0   66.8M  1 loop /snap/core24/1643
loop11       7:11     0   580K  1 loop /snap/snapd-desktop-integration/361
loop12       7:12     0   16.5M  1 loop /snap/firmware-updater/226
loop13       7:13     0    74M  1 loop /snap/core22/2411
sda          8:0      0   40G  0 disk
├─sda1       8:1      0     1M  0 part
└─sda2       8:2      0   40G  0 part /
sdb          8:16     0   10G  0 disk
└─sdb1       8:17     0   10G  0 part
ubuntu-01@ubuntu-01:~$

```

Step 3: Encrypt the partition

The command "luksFormat" is to create the encryption layer. This is the passphrase that needs to be entered by anyone who wants to access the device. Run "luksFormat" command on the newly created partition /dev/sdb1.

- \$ sudo cryptsetup -y -v luksFormat /dev/sdb1 crypt

```

Command failed with code 1 (wrong or missing parameters):
ubuntu-01@ubuntu-01:~$ sudo cryptsetup -y -v luksFormat /dev/sdb1
WARNING: Device /dev/sdb1 already contains a 'crypto_LUKS' superblock signature.

WARNING!
=====
This will overwrite data on /dev/sdb1 irrevocably.

Are you sure? (Type 'yes' in capital letters): YES
Enter passphrase for /dev/sdb1:
Verify passphrase:
Existing 'crypto_LUKS' superblock signature on device /dev/sdb1 will be wiped.
Existing 'crypto_LUKS' superblock signature on device /dev/sdb1 will be wiped.

Key slot 0 created.
Command successful.
ubuntu-01@ubuntu-01:~$

```


Step 4: Map the partition

A command "luksOpen" that brings a different level where we work with the encrypted device. So this will create a new device, and this new device is managed by the device mapper, so let's call it /dev/mapper/ CND.

- \$ sudo cryptsetup -v luksOpen /dev/sdb1 crypt

```
ubuntu-01@ubuntu-01:~$ sudo cryptsetup -y -v luksOpen /dev/sdb1 crypt
No usable token is available.
Enter passphrase for /dev/sdb1:
Verify passphrase:
Key slot 0 unlocked.
Command successful.
ubuntu-01@ubuntu-01:~$
```

Step 5: Format the partition

Create a filesystem on LUKS device. Run the following command:

- \$ sudo mkfs.ext4 /dev/mapper/crypt

```
ubuntu-01@ubuntu-01:~$ sudo mkfs.ext4 /dev/mapper/crypt
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 2617088 4k blocks and 655360 inodes
Filesystem UUID: 9f8f6991-7b69-42f8-b786-bbb8d9a247c4
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

Create a separate directory using the command:

- \$ sudo mkdir -p /mnt/encrypted_CND

```
ubuntu-01@ubuntu-01:~$ sudo mkdir -p /mnt/encrypted_CND
```

Verification:

ls -la /mnt/

#should show the encrypted_CND directory

```
ubuntu-01@ubuntu-01:~$ ls -la /mnt/
total 12
drwxr-xr-x  3 root root 4096 Jun  1 19:33 .
drwxr-xr-x 23 root root 4096 May 31 15:43 ..
drwxr-xr-x  2 root root 4096 Jun  1 19:33 encrypted_CND
ubuntu-01@ubuntu-01:~$
```

Step 6: Mount the partition

Mount LUKS device to the newly created directory using the command:

- `$ sudo mount -v /dev/mapper/crypt /mnt/encrypted_CND`

```
ubuntu-01@ubuntu-01:~$ sudo mount -v /dev/mapper/crypt /mnt/encrypted_CND
mount: /dev/mapper/crypt mounted on /mnt/encrypted_CND.
ubuntu-01@ubuntu-01:~$
```

To relabel and remount the new device, install necessary packages required: "policycoreutils". Install package using the command:

- `$ sudo apt install -y policycoreutils`

```
mount: /dev/mapper/crypt mounted on /mnt/encrypted_CND.
ubuntu-01@ubuntu-01:~$ sudo apt install -y policycoreutils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  selinux-utils
The following NEW packages will be installed:
  policycoreutils selinux-utils
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 261 kB of archives.
After this operation, 1,587 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/universe amd64 selinux-uti
64 3.5-2ubuntu2.1 [108 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 policycoreutils amd
-2 [153 kB]
Fetched 261 kB in 2s (121 kB/s)
Selecting previously unselected package selinux-utils.
(Reading database ...
```

To change the default context under SELinux, use "restorecon" command here. Run the following command:

- `restorecon -WRF /mnt/encrypted_CND`

```
Setting up policycoreutils (3.5-2) ...
Processing triggers for man-db (2.12.0-4build2) ...
ubuntu-01@ubuntu-01:~$ restorecon -WRF /mnt/encrypted_CND
ubuntu-01@ubuntu-01:~$
```

To mount the partition once again, run the following command:

- `$ sudo mount -v -o remount /mnt/encrypted_CND`

```
ubuntu-01@ubuntu-01:~$ sudo mount -v -o remount /mnt/encrypted_CND
mount: /dev/mapper/crypt mounted on /mnt/encrypted_CND.
ubuntu-01@ubuntu-01:~$
```

Verification:

`mount | grep encrypted_CND` #Check mount status

`ls -lZ /mnt/encrypted_CND` #Check show SELinux context applied

```
ubuntu-01@ubuntu-01:~$ mount | grep encrypted_CND
/dev/mapper/crypt on /mnt/encrypted_CND type ext4 (rw,relatime)
ubuntu-01@ubuntu-01:~$ ls -lZ /mnt/encrypted_CND
total 16
drwx----- 2 root root ? 16384 Jun  1 19:31 lost+found
```

Step 7: Retrieving LUKS details

Dump the details of the partition. Run the following command:

- \$ sudo cryptsetup luksDump /dev/sdb1

```
ubuntu-01@ubuntu-01:~$ sudo cryptsetup luksDump /dev/sdb1
LUKS header information
Version:          2
Epoch:           3
Metadata area:    16384 [bytes]
Keyslots area:    16744448 [bytes]
UUID:             16026d63-1962-4e18-aa1d-125a34c29122
Label:            (no label)
Subsystem:        (no subsystem)
Flags:            (no flags)

Data segments:
 0: crypt
   offset: 16777216 [bytes]
   length: (whole device)
   cipher: aes-xts-plain64
   sector: 512 [bytes]

Keyslots:
 0: luks2
   Key:          512 bits
   Priority:      normal
```

```
ubuntu-01@ubuntu-01: ~
Cipher:          aes-xts-plain64
Cipher key: 512 bits
PBKDF:           argon2id
Time cost:       4
Memory:          804017
Threads:         2
Salt:            91 6e ec fc cf cb 8a 16 2f 13 68 6b d4 af 5e 4f
                  15 43 89 8c 26 20 d4 ef 94 b5 6c 44 6b ac 25 e5
AF stripes: 4000
AF hash:         sha256
Area offset: 32768 [bytes]
Area length: 258048 [bytes]
Digest ID: 0

Tokens:
Digests:
 0: pbkdf2
   Hash:          sha256
   Iterations: 134432
   Salt:          98 57 46 b3 95 61 e9 c0 9d e7 4b 81 86 ef a1 f1
                  74 58 7f 4d ec a9 5a 9c d5 ad fe 0a 2e f0 f9 e4
   Digest:        1a bb 70 84 c1 1d 99 54 21 6b 62 6d 9e 9f 25 f1
                  db 3e f8 70 3f 6e 0b 0c a9 59 a5 95 87 d3 71 55
Show Apps ubuntu-01:~$
```

Step 8: Add a new LUKS key

Create a directory to store the new key. Run the following command to save the key inside the newly created directory. Run the following command:

- \$ sudo mkdir /etc/luks-keys
- \$ sudo chmod 700 /etc/luks-keys
- \$ sudo dd if=/dev/random of=/etc/luks-keys/luks-key bs=32 count=1

```
db 3e f8 70 3f 6e 0b 0c a9 59 a5 95 87 d3 71 55
ubuntu-01@ubuntu-01:~$ sudo mkdir /etc/luks-keys
ubuntu-01@ubuntu-01:~$ sudo chmod 700 /etc/luks-keys
```

```
ubuntu-01@ubuntu-01:~$ sudo dd if=/dev/random of=/etc/luks-keys/luks-key bs=32 count=1
1+0 records in
1+0 records out
32 bytes copied, 0.0017573 s, 18.2 kB/s
ubuntu-01@ubuntu-01:~$
```

To add new LUKS key using cryptsetup, run the following command:

- cryptsetup luksAddKey /dev/sdb1 /etc/luks-keys/luks-key -S 0

```
ubuntu-01@ubuntu-01:~$ sudo cryptsetup luksAddKey /dev/sdb1 /etc/luks-keys/luks-key -S 1
WARNING: The --key-slot parameter is used for new keyslot number.
Enter any existing passphrase:
ubuntu-01@ubuntu-01:~$ sudo cryptsetup luksDump /dev/sdb1 | grep -i "key slot"
```

```
ubuntu-01@ubuntu-01:~$ sudo stat /etc/luks-keys/luks-key
File: /etc/luks-keys/luks-key
Size: 32          Blocks: 8          IO Block: 4096   regular file
Device: 8,2      Inode: 921         Links: 1
Access: (0644/-rw-r--r--)  Uid: (  0/   root)   Gid: (  0/   root)
Access: 2026-06-01 20:01:16.582598826 +0530
Modify: 2026-06-01 20:01:16.582598826 +0530
Change: 2026-06-01 20:01:16.582598826 +0530
Birth: 2026-06-01 20:01:16.582598826 +0530
```

Verify the device is mapped and mounted

\$ sudo cryptsetup luksDump /dev/sdb1 | grep -E "^[[:space:]]*[0-9]+: luks2"

```
ubuntu-01@ubuntu-01:~$ sudo cryptsetup luksDump /dev/sdb1 | grep -E "^[[:space:]]*[0-9]+: luks2"
0: luks2
1: luks2
ubuntu-01@ubuntu-01:~$
```

```
ubuntu-01@ubuntu-01:~$ lsblk /dev/sdb
NAME        MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sdb          8:16   0    10G  0 disk
└─sdb1       8:17   0    10G  0 part
   └─crypt    252:0   0    10G  0 crypt /mnt/encrypted_CND
ubuntu-01@ubuntu-01:~$
```

Verify disk space is accessible

\$ mount | grep encrypted_CND

\$ df -h /mnt/encrypted_CND

```
ubuntu-01@ubuntu-01:~$ mount | grep encrypted_CND
/dev/mapper/crypt on /mnt/encrypted_CND type ext4 (rw,relatime)
ubuntu-01@ubuntu-01:~$ df -h /mnt/encrypted_CND
Filesystem      Size  Used Avail Use% Mounted on
/dev/mapper/crypt 9.8G   24K   9.3G   1% /mnt/encrypted_CND
ubuntu-01@ubuntu-01:~$
```

Verify LUKS2 slots (already confirmed, quick recheck)

\$ sudo cryptsetup luksDump /dev/sdb1 | grep -E "^[[:space:]]*[0-9]+: luks2"

```
ubuntu-01@ubuntu-01:~$ sudo cryptsetup luksDump /dev/sdb1 | grep -E "^[[:space:]]*[0-9]+: luks2"
: luks2"
[sudo] password for ubuntu-01:
 0: luks2
 1: luks2
ubuntu-01@ubuntu-01:~$
```

Verify key file unlocks the device

\$ sudo cryptsetup luksOpen --test-passphrase /dev/sdb1 --key-file /etc/luks-keys/luks-key && echo "Key file OK"

```
ubuntu-01@ubuntu-01:~$ sudo cryptsetup luksOpen --test-passphrase /dev/sdb1 --key-file /etc/luks-keys/luks-key && echo "Key file OK"
Key file OK
ubuntu-01@ubuntu-01:~$
```

Verify you can write and read data

\$ sudo touch /mnt/encrypted_CND/testfile.txt

\$ sudo ls -la /mnt/encrypted_CND/

```
ubuntu-01@ubuntu-01:~$ sudo touch /mnt/encrypted_CND/testfile.txt
ubuntu-01@ubuntu-01:~$ sudo ls -la /mnt/encrypted_CND/
total 24
drwxr-xr-x 3 root root 4096 Jun  1 20:56 .
drwxr-xr-x 3 root root 4096 Jun  1 19:33 ..
drwx----- 2 root root 16384 Jun  1 19:31 lost+found
-rw-r--r-- 1 root root    0 Jun  1 20:56 testfile.txt
ubuntu-01@ubuntu-01:~$
```

Full unmount and remount test (the real proof)

Unmount

\$ sudo umount /mnt/encrypted_CND

\$ sudo cryptsetup luksClose crypt

```
ubuntu-01@ubuntu-01:~$ sudo umount /mnt/encrypted_CND
ubuntu-01@ubuntu-01:~$ sudo cryptsetup luksClose crypt
```

Reopen with key file

\$ sudo cryptsetup luksOpen /dev/sdb1 crypt --key-file /etc/luks-keys/luks-key

Remount

\$ sudo mount /dev/mapper/crypt /mnt/encrypted_CND

Confirm file survived

\$ sudo ls -la /mnt/encrypted_CND/


```

key file: unknown option
ubuntu-01@ubuntu-01:~$ sudo cryptsetup luksOpen /dev/sdb1 crypt --key-file /etc/luks-key
s/luks-key
ubuntu-01@ubuntu-01:~$ sudo mount /dev/mapper/crypt /mnt/encrypted_CND
ubuntu-01@ubuntu-01:~$ sudo ls -la /mnt/encrypted_CND/
total 24
drwxr-xr-x 3 root root 4096 Jun  1 20:56 .
drwxr-xr-x 3 root root 4096 Jun  1 19:33 ..
drwx----- 2 root root 16384 Jun  1 19:31 lost+found
-rw-r--r-- 1 root root    0 Jun  1 20:56 testfile.txt
ubuntu-01@ubuntu-01:~$

```

EXPECTED OUTPUT

```

root@ubuntu-01:/home/ubuntu-01# sudo mount /dev/mapper/crypt /mnt/encrypted_CND
root@ubuntu-01:/home/ubuntu-01# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
loop0        7:0    0     4K  1 loop  /snap/bare/5
loop1        7:1    0    74M  1 loop  /snap/core22/2292
loop2        7:2    0   66.8M  1 loop  /snap/core24/1643
loop3        7:3    0    74M  1 loop  /snap/core22/2411
loop4        7:4    0   16.5M  1 loop  /snap/firmware-updater/226
loop5        7:5    0   18.5M  1 loop  /snap/firmware-updater/210
loop6        7:6    0  531.4M  1 loop  /snap/gnome-42-2204/247
loop7        7:7    0  251.7M  1 loop  /snap/firefox/7766
loop8        7:8    0   91.7M  1 loop  /snap/gtk-common-themes/1535
loop9        7:9    0   10.8M  1 loop  /snap/snap-store/1270
loop10       7:10   0   48.1M  1 loop  /snap/snapd/25935
loop11       7:11   0   49.3M  1 loop  /snap/snapd/26865
loop12       7:12   0    576K  1 loop  /snap/snapd-desktop-integration/343
loop13       7:13   0    580K  1 loop  /snap/snapd-desktop-integration/361
sda          8:0    0   40G   0 disk
├─sda1       8:1    0     1M   0 part
└─sda2       8:2    0   40G   0 part  /
sdb          8:16   0   10G   0 disk
└─sdb1       8:17   0   10G   0 part
   └─crypt   252:0    0   10G   0 crypt  /mnt/encrypted_CND
root@ubuntu-01:/home/ubuntu-01#

```

```

root@ubuntu-01:/home/ubuntu-01# dd if=/dev/random of=/etc/luks-keys/CND_key bs=32 count=
1
1+0 records in
1+0 records out
32 bytes copied, 8.0746e-05 s, 396 kB/s
root@ubuntu-01:/home/ubuntu-01#

```